What Can a Small Human Audit Establish About a Leaderboard Ranking? Two Verifiers and the Assumption That Does the Work

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Abstract

Math leaderboards score free-text answers with automated verifiers and check the verifiers with small human audits. We ask what such an audit can establish about a *ranking* of models, on the MATH-Hard subset of the Open LLM Leaderboard (17 models, 136 pairwise orderings, and a 400-item human audit). The harness runs two verifiers; they disagree on 3.63% of 22,508 responses and reverse 8 orderings, 7 of them clear under both verifiers' sampling error. Used the standard way—one verifier, error rates estimated per model—the audit resolves 0 orderings at 95% coverage. The audit also shows where verifier error concentrates: on responses where the two verifiers agree, the transcribed human answer disagrees with them on 1 of 287 (95% upper bound 1.9%), while the model-specific errors it finds lie where they disagree. If agreement between the verifiers is equally reliable for every model, the same audit resolves 84 of the 118 orderings that sampling error permits. That assumption does most of the work: allowing each model's agreement error to vary within twice the pooled bound leaves 43, within five times 7, and dropping the assumption leaves 0; reaching 84 without it would take on the order of 27,200 labels. We therefore recommend stratifying audits on joint verifier outcome, stating the cross-model homogeneity assumption, and spending part of the audit on testing it; we give a preregistered 458-item design. Results are on one benchmark, and the audit validates answer extraction, not semantic equivalence.

1 Introduction

A leaderboard number for a math benchmark is the output of a verifier: code that extracts a final answer from free text and compares it with a reference. Verifiers fail in known ways—unboxed answers, notation they cannot normalize, text after the answer—and such failures led the Open LLM Leaderboard to rescore its models with a new verifier [Kydlicek et al., 2025]. The usual safeguard is a human audit of a few hundred responses, summarized as an error rate. Leaderboards exist to order models, however, and an error rate that is small on average can still reverse an ordering if it falls unevenly on one model.

We ask what a small audit can establish about orderings, using public MATH-Hard generations from 17 models, the two verifiers the evaluation harness runs on every response, and a 400-item human audit. We say an ordering is *resolved* when an interval for the difference in true accuracy excludes zero at 95% coverage under the stated assumptions.

Preprint.

Verifier choice reverses orderings (Section 5.1). Ranking by one verifier or the other reverses 8 of 136 orderings; 7 are clear under both verifiers' sampling error.

Per-model auditing with one verifier resolves nothing (Section 5.2). The verifier is accurate on the audited items, but each model contributes a handful of them, and the audit resolves 0 orderings.

Verifier agreement concentrates the observed error (Section 5.3). Where the two verifiers agree, the transcribed human answer disagrees with them on 1 of 287 audited responses; the model-specific errors the audit finds are in the disagreements.

What agreement buys depends on one assumption (Section 5.4). If agreement is equally reliable across models, the existing audit resolves 84 of the 118 orderings sampling error permits. The count falls to 43 if each model's agreement error may be twice the pooled bound, and to 0 without the assumption; the audit is too small to test it model by model. The practical contribution is to make this assumption explicit, quantify how much the ranking depends on it, and design the next audit to test it (Section 5.5).

The estimators are classical [Neyman, 1938, Begg and Greenes, 1983]; the contributions are the measurements, the sensitivity analysis, and the design they imply.

2 Related Work

Two-phase sampling and verification bias. Classifying everyone cheaply and verifying a subsample stratified by the cheap result is two-phase sampling [Neyman, 1938]; Tenenbein [1970] treats misclassified binomial data including phase sizes, and Begg and Greenes [1983] give the correction when verification depends only on the test result. Our estimators are these. When the reference is instead collected on units of known true status, sensitivity and specificity are the natural parameters [Lang and Reiczigel, 2014].

Hybrid human–automatic evaluation. Prediction-powered inference [Angelopoulos et al., 2023] combines automatic labels with few human ones; active [Zrnic and Candès, 2024] and stratified [Fisch et al., 2024] variants choose what to label, the latter for language-model evaluation. Boyeau et al. [2025] apply the approach to model evaluation and Chatzi et al. [2024] to rank sets of LLMs from pairwise preferences. Lee et al. [2026] correct one model's LLM-judge accuracy with sensitivity and specificity estimated on calibration items drawn by true label, and allocate those items between the two labels. Relative to these, we study many models and all their pairwise orderings, allow error to differ by model, and use agreement between two deterministic verifiers as the stratifier.

Verifier reliability. VerifyBench [Li et al., 2025] and CompassVerifier [Liu et al., 2025] compare verifiers' accuracy; Kydlicek et al. [2025] document the extraction failures behind the Open LLM Leaderboard [Fourrier et al., 2024] rescoring; Biderman et al. [2024] discuss evaluation reproducibility.

Concurrent work sharing data. Pall [2026] (same author, under review) uses the same audit and roster to bound accuracy by a verifier's parse-failure count; its estimand and questions differ.

3 Estimand and Identification

Reference. For model m and item i, a human transcribes the final answer the response commits to; $Y^*_{mi} = 1$ if that transcription is equivalent to the reference answer under the harness's normalization and equivalence routine, and $Y^*_{mi} = 0$ otherwise, including when the response commits to no answer. The reference therefore validates *answer extraction*; equivalence errors shared by the verifiers and the routine are not identified (Section 4). The target is $A^*_m = P(Y^* = 1)$ over the benchmark's items.

One verifier. With verifier decision $\hat{Y}$ and accuracy $a_m = P(\hat{Y} = 1)$, known because the verifier runs on every item, an audit stratified on $\hat{Y}$ estimates $q_m = P(Y^* = 0 \mid \hat{Y} = 1)$ and

$r_m = P(Y^* = 1 \mid \hat{Y} = 0)$, and

$$A_m^* = a_m(1 - q_m) + (1 - a_m)\, r_m. \tag{1}$$

The error rates $\alpha_m = P(\hat{Y} = 1 \mid Y^* = 0) = a_m q_m/(1 - A_m^*)$ and $\beta_m = P(\hat{Y} = 0 \mid Y^* = 1) = (1 - a_m)r_m/A_m^*$ follow by Bayes' rule; feeding q_m, r_m directly into the sensitivity/specificity correction $(a - \alpha)/(1 - \alpha - \beta)$ is biased and can return empty intervals (Appendix D).

Two verifiers. Each response falls in a joint cell $c \in \{11, 10, 01, 00\}$ (first index `score_boxed`, second `exact_match`; 1 = credit). With census shares s_{mc},

$$A_m^* = \sum_c s_{mc}\, p_{mc}, \qquad p_{mc} = P(Y^* = 1 \mid \text{cell } c,\ m). \tag{2}$$

$\Delta^* = A_1^* - A_2^*$ is linear in the p_{mc}, so over a box of values its range is attained at corners and is sharp.

Assumptions compared. Disagreement cells are always left model-specific: $p_{m,10}, p_{m,01} \in [0, 1]$ unless audited. For the concordant cells we compare

- *homogeneity*: $p_{m,11} = p_{11}$, $p_{m,00} = p_{00}$ for all models, estimated from the pooled audit;
- *bounded heterogeneity* $L \geq 1$: each model's $1 - p_{m,11}$ and $p_{m,00}$ lie anywhere in $[0,\ L \cdot u]$, independently across models, where u is the pooled upper confidence bound;
- *no pooling*: each model's concordant rates from its own audited items only.

Pooling a single verifier's (q, r) across models would instead give $\Delta^* = (a_1 - a_2)(1 - q - r)$, whose sign never changes: it assumes away the model-specific error that can reorder models.

Coverage. For each ordering we split the 5% error budget between the audit (2.5%) and benchmark sampling (2.5%) and combine them by the union bound, which needs no independence. Within the audit part, each of the k estimated parameters the ordering depends on gets a Clopper–Pearson interval at level $0.975^{1/k}$; the estimates come from disjoint samples, so they hold jointly with probability at least 0.975. Parameters bounded logically ($[0, 1]$) cost no coverage. The sampling part is a normal interval for the difference in verifier accuracy. Details and a simultaneous-over-all-orderings variant are in Appendix B.

4 Data and Audit

Benchmark and verifiers. MATH-Hard [Hendrycks et al., 2021] on the Open LLM Leaderboard v2 [Fourrier et al., 2024]: 1,324 items in 7 subjects with public per-item generations. The harness scores every response with `exact_match`, which extracts the answer after a Minerva-style "final answer" phrase [Lewkowycz et al., 2022] and is the leaderboard's reported metric, and `score_boxed`, which extracts the last `\boxed{}`. Both then apply the same normalization and equivalence routine to the reference.

Human audit. 400 (model, item) responses were drawn from 28 models' pooled responses, stratified by `score_boxed` outcome × subject × response-length quartile. Labelers saw the problem and response only and transcribed the final answer the response commits to, or marked it as committing to none (`NONE`) or to two unresolved answers (`CONTRADICTORY`). The transcription is compared with the reference by the verifiers' own equivalence routine, so the audit measures extraction error; an equivalence error shared by the verifiers and the reference routine is invisible to it. Three labelers overlapped on 151 and 75 items (Cohen's κ 0.61, 0.82, 0.79 across pairs).

Models and cells. We use the 17 models with audited items in both `score_boxed` strata. Their audited responses fall in the joint cells as in Table 2. Because inclusion depends only on the `score_boxed` stratum, subject and length, it is independent of the `exact_match` outcome within a stratum, so the joint cells are valid domains of the existing design.

Table 1: Orderings that reverse between the two verifiers. "Sampling-clear": the accuracy difference exceeds 1.96 standard errors under the named verifier(s). Last column: audited responses credited only by `score_boxed` whose transcribed answer is correct, for each model.

Pair	higher by score_boxed	higher by exact_match	sampling-clear	audited sb-only correct/n
Yi-1.5-9B-Chat vs zephyr-orpo-141b-A35b-v0.1	Yi-1.5-9B-Chat	zephyr-orpo-141b-A35b-v0.1	both	1/9 ; 2/2
Yi-1.5-9B-Chat vs Qwen2-7B	Yi-1.5-9B-Chat	Qwen2-7B	em only	1/9 ; 2/2
c4ai-command-r-v01 vs gpt-j-6b	c4ai-command-r-v01	gpt-j-6b	both	2/6 ; 0/0
c4ai-command-r-v01 vs phi-2	c4ai-command-r-v01	phi-2	both	2/6 ; 0/0
c4ai-command-r-v01 vs Mistral-7B-v0.1	c4ai-command-r-v01	Mistral-7B-v0.1	both	2/6 ; 0/0
c4ai-command-r-v01 vs mpt-7b	c4ai-command-r-v01	mpt-7b	both	2/6 ; 0/0
c4ai-command-r-v01 vs falcon-40b	c4ai-command-r-v01	falcon-40b	both	2/6 ; 0/0
c4ai-command-r-v01 vs falcon-7b	c4ai-command-r-v01	falcon-7b	both	2/6 ; 0/1

Table 2: Audited responses of the 17 models by joint verifier cell, and how many have a transcribed answer equal to the reference (responses committing to no answer count as not equal).

Cell	both credit	both reject	`score_boxed` only	`exact_match` only
Audited	142	145	32	1
Transcription correct	141	0	16	0

5 Results

5.1 Verifier choice reverses orderings

The two verifiers disagree on 817 of 22,508 responses (3.63%, 95% CI [3.39%, 3.88%]; 494 credited only by `exact_match`, 323 only by `score_boxed`), from 0.60% to 9.59% per model. Table 1 lists the 8 orderings that reverse between them. 6 involve `c4ai-command-r-v01`, which `exact_match` scores at 0. The audit contains few items per model, but in each case its evidence points the same way: `command-r-v01`'s responses credited only by `score_boxed` are mostly wrong, as are `Yi-1.5-9B-Chat`'s (most commit to no answer), while `zephyr-orpo` and `Qwen2-7B`'s are correct.

5.2 Per-model correction with one verifier resolves nothing

With `score_boxed` and each model's own audited items, the audit finds 5 false credits in 162 credited items and 0 false rejections in 127 rejected ones, and corrected point estimates reorder only 2 pairs. Yet at 95% coverage the audit resolves 0 orderings, against 118 that sampling error alone permits (Appendix E, Figure 3). 94% of each model's interval width comes from the rejected stratum: zero observed errors in 1–17 items bounds r_m only loosely, and the rejected stratum holds most of each model's responses. Simulated single-verifier audits resolve half the orderings at about 2,312 labels (Appendix E). Plug-in Neyman allocation resolves none at any budget, because zero-event pilot strata receive no labels.

5.3 Where verifier error concentrates

Where the verifiers agree, the transcribed answer disagrees with them on 1 of 287 audited responses; the exact 95% upper bound on this rate is 1.9% (3.9% and 2.5% for the two concordant cells separately). This is a low observed rate with a finite-sample bound, not proof that agreement is reliable, and it says nothing about whether the rate is the same for every model. Where `score_boxed` credits and `exact_match` does not, 16 of 32 transcriptions are correct (Table 2); every false credit of `command-r-v01` is in this cell. The `exact_match`-only cell has 493 responses in the population but 1 audited, because the audit was stratified on `score_boxed`.

5.4 What verifier agreement buys, and what it assumes

Under homogeneity, with disagreement cells left model-specific and bounded only by $[0, 1]$, the existing audit resolves 84 of the 118 orderings sampling permits (101 of 136 from the audit alone; 76 of 103 with coverage simultaneous over all orderings; 84 if responses without a committed answer

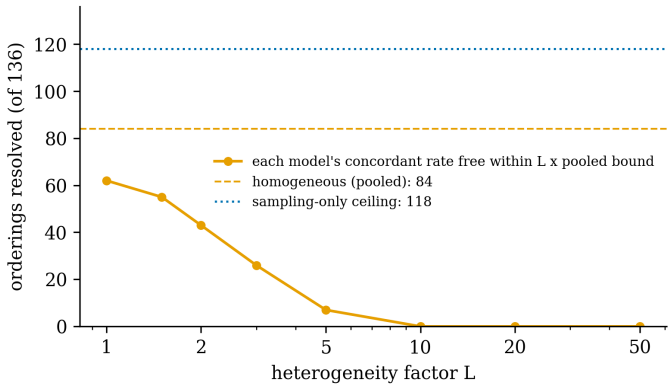


Figure 1: Orderings resolved by the existing audit, at 95% coverage per ordering, as the concordant-cell assumption is relaxed. Dashed: concordant error homogeneous across models. Solid: each model's concordant error free within L times the pooled bound. Dotted: orderings that sampling error alone permits.

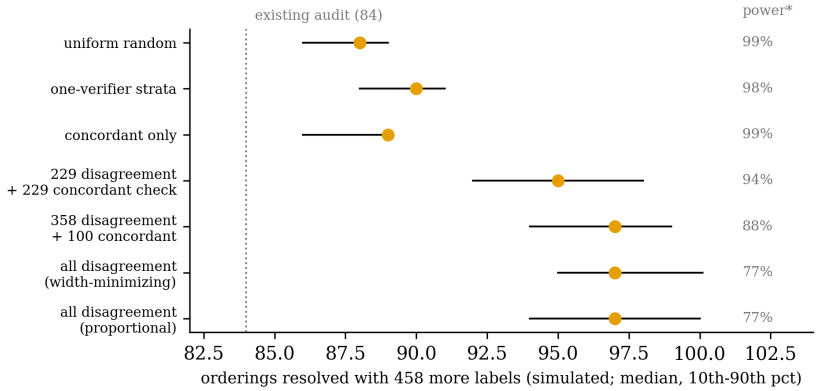


Figure 2: Projected, not observed. Orderings resolved (homogeneous analysis) after 458 further labels under each design; 200 simulated audits per design, truths drawn from the existing audit's posterior; median and 10th–90th percentile. Right column (*): median across models of the probability of observing at least one concordant error for a model whose concordant error rate is 10%.

are dropped rather than counted as incorrect). Figure 1 shows how much of this is the assumption. If each model's concordant error may lie anywhere within the pooled bound ($L = 1$), 62 orderings remain; at $L = 2$, 43; at $L = 5$, 7; at $L = 10$, 0. With no pooling at all the count is 0. Recovering 84 without the assumption would require about 27,200 concordant labels (57 at 3,400), because every model's two concordant cells, which hold most of its responses, would need their own tight bounds.

The assumption is plausible on mechanism: two extractors that read answers in different ways and return the same verdict rarely both err on the same response. The audit cannot confirm it model by model: models have 0 to 37 audited items per concordant cell.

5.5 Designing the next audit

Figure 2 compares designs for 458 further labels under the same objective. Under homogeneity, spending most labels on disagreements gives a median of 95–97 resolved orderings across truth scenarios, against 88 for uniform sampling and 90 for one-verifier strata; allocations within the disagreement cells differ by at most a few orderings, so none is uniquely best. The same designs differ sharply in how well they test the assumption the result rests on: an all-disagreement design would

detect a model with a 10% concordant error rate with median probability 77% (worst model 47%), a half-and-half design with 94% (77%) while still reaching 94–95. We preregistered the half-and-half design: 229 disagreement responses stratified by model and 229 concordant responses spread evenly over models. These results are projections from simulations that draw truths from the same audit used to choose the design; they are not observed outcomes.

6 Recommendations

1. **Run two verifiers and report their disagreement.** It costs nothing, it reverses orderings, and it locates the errors a small audit can find.
2. **Stratify the audit on joint verifier outcome and state the homogeneity assumption.** A ranking claim from a small audit rests on treating verifier agreement as equally reliable across models; say so, and report how the claim changes as the assumption is relaxed (Figure 1).
3. **Spend part of the audit testing that assumption.** Labels on concordant responses spread over models are what can falsify it; labels on disagreements are what resolve orderings once it holds.
4. **Do not pool one verifier's error rates across models to support a ranking**, and match the estimator to the sampling: an audit drawn within verifier strata estimates q and r (Eq. (1)).
5. **Say what the audit measures.** Transcription audits validate extraction; semantic equivalence needs a separate check.

7 Limitations

1. **The headline count rests on an untested assumption.** 84 resolved orderings require concordant error to be homogeneous across models; the existing audit supports a low pooled rate but cannot test homogeneity, and the count falls quickly as it is relaxed.
2. **One benchmark and an older roster.** All results are on MATH-Hard with Open LLM Leaderboard v2 models. We did not find a second benchmark with two independent live verifiers and an existing human audit; the methodology applies wherever two verifiers can be run, but its payoff depends on how concentrated verifier error is there.
3. **The reference is transcription plus automated comparison**, with inter-labeler κ of 0.61–0.82. The two verifiers share normalization code, so their agreement is not independent evidence about equivalence errors.
4. **The `exact_match`-only cell is barely audited** (1 item) and is bounded by $[0, 1]$ per model rather than estimated.
5. **Design results are projections**, from simulations whose truths come from the same audit that motivated the design.

8 Conclusion

A small human audit of one verifier, read per model, resolves no MATH-Hard ordering. Read through the agreement of two verifiers the harness already runs, it resolves 84 of the 118 orderings sampling permits, but only if agreement is equally reliable across models; without that assumption, resolving as many would take on the order of 27,200 labels. The assumption is where a leaderboard's ranking claim now lives, and the next audit should be designed to test it.

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A Derivations

Identities. Conditioning on the verifier outcome, $P(Y^* = 1) = P(Y^* = 1 \mid \hat{Y} = 1)P(\hat{Y} = 1) + P(Y^* = 1 \mid \hat{Y} = 0)P(\hat{Y} = 0) = a(1-q) + (1-a)r$; conditioning on the joint cell gives Eq. (2). Bayes' rule gives $\alpha = aq/(1-A^*)$ and $\beta = (1-a)r/A^*$; substituting these into $(a-\alpha)/(1-\alpha-\beta)$ recovers A^* exactly (checked symbolically in the released code), substituting q, r directly does not.

Sharpness. $\Delta^* = \sum_j k_j \theta_j + \text{const}$ with coefficients k_j given by census shares ($s_{1c} - s_{2c}$ for a pooled parameter; s_{1c} and $-s_{2c}$ for model-specific ones). Over a box $\prod_j [\theta_j^-, \theta_j^+]$ the minimum takes θ_j^- where $k_j > 0$ and θ_j^+ where $k_j < 0$, the maximum the reverse; both are attained, so the range is sharp.

Pooling one verifier's rates. With common (q, r), $\Delta^* = (a_1 - a_2)(1 - q - r)$, whose sign is that of $a_1 - a_2$ whenever $q + r < 1$.

B Coverage

Fix an ordering $(1, 2)$. Let E_{aud} be the event that every estimated parameter it depends on lies in its Clopper–Pearson interval, and E_{samp} the event that the benchmark-sampling interval for the accuracy difference covers its target. The k estimated parameters are proportions from disjoint audited samples (different cells, or different models' items), so with per-parameter level $0.975^{1/k}$, $P(E_{\text{aud}}) \geq 0.975$ (Clopper–Pearson intervals are conservative, so the bound holds exactly, not asymptotically). $P(E_{\text{samp}}) \approx 0.975$ from the normal approximation for a difference of two proportions over 1,324 items. On E_{aud}, Δ^* lies in the projected range (sharpness above); on $E_{\text{aud}} \cap E_{\text{samp}}$, the widened interval covers the superpopulation difference. By the union bound $P(E_{\text{aud}} \cap E_{\text{samp}}) \geq 1 - 0.025 - 0.025 = 0.95$, with no independence assumption between the audit sample and the benchmark items. Parameters bounded logically ($[0, 1]$ for unaudited disagreement cells, or the stated bound under bounded heterogeneity) are not estimated and use no coverage. Under homogeneity the audit part depends only on the two pooled parameters and so holds for all orderings at once; the simultaneous variant reported in Section 5.4 additionally puts the sampling part at $0.025/136$ per ordering. The coverage is conditional on the stated assumption (homogeneity or bounded heterogeneity); it does not account for the assumption being false.

C Why full disagreement auditing stops short of the ceiling

If every disagreement response were audited and each model's disagreement rates equalled the pooled audited value, 110 of the 118 sampling-permitted orderings would be resolved. 9 sampling-permitted orderings fail, and one ordering that sampling alone does not resolve becomes resolved. The failures are not for lack of audit information: correcting the disagreement mass shrinks the accuracy gap below sampling error, so the pair's `score_boxed` gap is significant but its corrected gap is not (5 of them involve `command-r-v01`, whose `score_boxed` accuracy is mostly disagreement mass). This projection fixes the disagreement rates at the pooled value and is illustrative only.

D Simulation: plug-in versus two-phase estimator

Finite benchmark of $N = 1{,}324$ items; verifier errors drawn with the stated (α, β); audit by simple random sampling within verifier strata (6–40 items per stratum); 4,000 replicates. "Plug-in" feeds the audited stratum rates into $(a - \alpha)/(1 - \alpha - \beta)$. Coverage is for the Šidák-adjusted Wilson projection used in this check.

Scenario (A^*, α, β)	bias, Eq. (1)	bias, plug-in	cover, Eq. (1)	cover, plug-in	empty, plug-in
typical (0.2, 0.01, 0.02)	-0.0001	-0.027	1.000	0.998	0.00
low accuracy (0.02, 0.01, 0.02)	+0.0002	-0.560	1.000	0.237	0.60
zero accuracy (0.0, 0.01, 0.0)	+0.0000	—	1.000	—	1.00
high error (0.4, 0.1, 0.15)	-0.0003	-0.078	1.000	0.961	0.00
perfect verifier (0.2, 0.0, 0.0)	+0.0000	+0.000	1.000	1.000	0.00

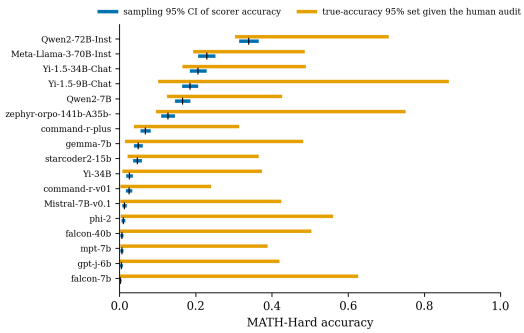


Figure 3: Per-model `score_boxed` accuracy: sampling 95% CI (blue) and true-accuracy set from the one-verifier, per-model audit (orange; Eq. (1), Šidák-adjusted Wilson intervals; illustrative).

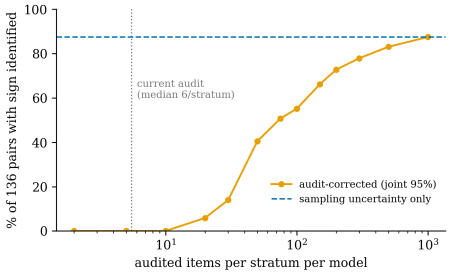


Figure 4: One verifier, per model (audit only, Wilson): share of orderings resolved as audited items per stratum per model grow, each model's observed error rates held fixed.

E One-verifier audit design

For a model with zero observed errors in n items per stratum the Wilson width is about $z^2/(n+z^2)$; equating it with the sampling width $2z\sqrt{a(1-a)/N}$ gives $n^* \approx (z/2)\sqrt{N/(a(1-a))} - z^2$, i.e. 71, 86 and 160 items per stratum per model at $a = 0.35, 0.20, 0.05$. Simulated fresh audits (200 replicates, four truth scenarios, 289 determinate labels scaled by 1–64) compare five allocation rules at equal budget; plug-in Neyman resolves nothing at any budget, and the rule minimizing summed zero-event widths ($n \propto \sqrt{a_m}$, $\sqrt{1-a_m}$) is best or tied. Curves: `reaudit_results.json`.

F Common-mechanism diagnostic

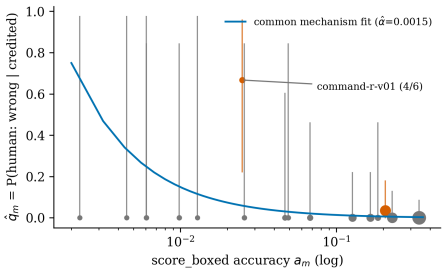


Figure 5: Observed $\hat{q}_m$ (exact 95% CI) against `score_boxed` accuracy, with the rate implied by one common error mechanism.

Under one (α, β) for all models, each model's (q_m, r_m) is implied by a_m. The grid maximum-likelihood fit gives $\hat{\alpha} = 0.0015$, $\hat{\beta} = 0$, and a likelihood ratio of 24.8 against per-model rates

(parametric bootstrap $p = 0.010$). `command-r-v01` contributes 15.3; leaving it out gives $p = 0.42$, leaving out any other model gives p between 0.002 and 0.026 (500 bootstrap replicates each). The rejection is one model, whose errors all lie in a disagreement cell; we report it as a diagnostic, not a general finding.

G Per-model audit, one verifier

Model	a_m	false cr./aud.	false rej./aud.	indet. (cr./rej.)	$\hat{A}^*_m$	95% set
falcon-7b	0.002	0/1	0/3	1/3	0.002	[0.00, 0.63]
gpt-j-6b	0.005	0/1	0/7	0/1	0.005	[0.00, 0.42]
mpt-7b	0.006	0/2	0/8	0/5	0.006	[0.00, 0.39]
falcon-40b	0.006	0/1	0/5	0/0	0.006	[0.00, 0.50]
phi-2	0.010	0/2	0/4	0/0	0.010	[0.00, 0.56]
Mistral-7B-v0.1	0.013	0/1	0/7	0/1	0.013	[0.00, 0.42]
c4ai-command-r-v01	0.025	4/6	0/17	0/3	0.008	[0.00, 0.24]
Yi-34B	0.026	0/2	0/9	0/0	0.026	[0.01, 0.37]
starcoder2-15b	0.047	0/4	0/10	0/1	0.047	[0.02, 0.36]
gemma-7b	0.049	0/2	0/6	0/1	0.049	[0.01, 0.48]
c4ai-command-r-plus	0.068	0/6	0/14	1/0	0.068	[0.04, 0.31]
zephyr-orpo-141b-A35b-v0.1	0.127	0/15	0/2	0/0	0.127	[0.10, 0.75]
Qwen2-7B	0.165	0/15	0/11	0/0	0.165	[0.12, 0.43]
Yi-1.5-9B-Chat	0.185	0/6	0/1	8/3	0.185	[0.10, 0.86]
Yi-1.5-34B-Chat	0.206	1/29	0/9	0/1	0.199	[0.16, 0.49]
Meta-Llama-3-70B-Instruct	0.229	0/27	0/10	0/0	0.229	[0.19, 0.49]
Qwen2-72B-Instruct	0.339	0/42	0/4	2/0	0.339	[0.30, 0.71]

H Closest prior work

Work	Estimand	Human sampling	Rank	Two verif.	Allocation
Two-phase / verification bias [Neyman, 1938, Tenenbein, 1970, Begg and Greenes, 1983]	prevalence, Se/Sp	on the cheap result	no	no	phase sizes
Lee et al. [2026]	one model's accuracy	by true label	no	no	between labels
Stratified PPI [Fisch et al., 2024]	a mean metric	stratified	no	no	across strata
PP-Ranking [Chatzi et al., 2024]	rank-sets, many models	random	yes	no	none
AutoEval [Boyeau et al., 2025]	accuracy, many models	random	partly	no	none
This paper	all pairwise orderings	joint-verifier strata	yes	yes	disagreement vs. check

The estimators here are those of the first row. What differs is the question (every pairwise ordering of many models), the use of two deterministic verifiers' agreement as the stratifier, the explicit cross-model homogeneity assumption and its sensitivity analysis, and an audit design that tests that assumption.

I Reproducibility

Every number in this paper is a macro generated by `scripts/fb_rebuild_numbers.py` from result files written by `fb_joint_verifier.py`, `fb_reaudit.py`, `fb_rebuild_analysis.py` and `fb_core_estimand_audit.py`; figures by `fb_rebuild_figures.py`. The estimator is in `flipbudget.pv`, with tests. The preregistered design for the next audit is `TC_EXPANSION_PREREGISTRATION.md`, Amendment 3, and its blinded sheet is `labeling/audit_v2_l1.csv`.